

img2threejs: one image to a production-ready interactive 3D asset

A quality-gated, code-only reconstruction workflow with bounded autonomy, preserved evidence and an explicit human handoff



INPUT / ONE REFERENCE

Visible art direction. Hidden geometry declared approximate.

THE IMG2THREEJS BOUNDED PIPELINE

STAGED RECONSTRUCTION, NOT ONE-SHOT MESH GENERATION

01

OBSERVE

Macro-to-micro image analysis; visible facts separated from hidden-side inference.

02

CONTRACT

Suitability, complexity, quality contract, detail inventory and PBR evidence.

03

SPEC

27-component hierarchy with materials, pivots, sockets and action anchors.

04

BUILD

Pass-locked TypeScript and Three.js generation with deterministic scaffolding.

05

REVIEW

Browser captures, comparison sheets, diagnostics and recorded correction decisions.

HARD STOP

blockout 3/3

HUMAN AUTHORIZATION

override + override v2

AUDIT RECORD

state remains stopped



OUTPUT / LIVE SYSTEM

Interactive, inspectable, animation-ready and GLB-exportable.

27/27

SPECIFIED PARTS BUILT

28

SEMANTIC NODES

3/3

BOUNDED LOOP HARD STOP

36/36

UNIT TESTS PASSED

3/3

BROWSER JOURNEYS PASSED

01 / PLUGIN OUTPUTS

What img2threejs produced

- Reference intake, suitability limits and macro-to-micro decomposition.
- PBR evidence, quality contract and a strict 27-component sculpt specification.
- Generated TypeScript scaffold, browser captures and comparison evidence.
- Persistent state, pass locks and review history for a reproducible audit trail.

02 / CONTROLLED HANDOFF

The guardrail actually fired

- After three blockout corrections, the workflow stopped at its configured limit.
- No state reset, silent bypass or false pass was used.
- Emmanuel explicitly authorized override, then override v2, with every v1 file preserved.
- The stopped state remains evidence of bounded autonomy, not a failed record to erase.

03 / PRODUCTION HANDOFF

What humans finished and proved

- Manual V2 refinement corrected the bow, pod, chains, mechanics, controls and UI.
- The runtime added selection, real pivots, two-button Fire/Load logic and GLB export.
- Strict coverage passed at 27/27 parts, plus 36/36 unit and 3/3 browser tests.
- GitHub and Railway turned the audited reconstruction into a public live product.

WHY THIS MATTERS TO A CTO

img2threejs made the process inspectable: specifications, evidence, pass state and correction limits survive the conversation. The plugin did not claim success when its gate failed. Human judgment resumed the work explicitly, while the production tests verify a different claim: that the shipped system works.

LIVE <https://ballista-production.up.railway.app>

SOURCE <https://github.com/edemaistre/steampunk-ballista-turret>

